

What Is Needed Before Market Release?

Production-readiness guide and release-gate checklist

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Product: NexusOne AI - Autonomous Enterprise Intelligence Platform

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KEY MESSAGE

Completing every feature phase means the roadmap is complete. Market release requires separate evidence that NexusOne AI is secure, reliable, scalable, supportable, legally usable, and valuable for a defined customer use case.

Use this document as a working release checklist. Replace each "TBD" with an accountable owner, target release, evidence link, and final sign-off.

Executive Summary

A market-ready release is not defined only by the number of completed modules. It is defined by validated customer value and controlled operational risk. NexusOne AI should pass the following release gates before public production use.

G1	Product scope	A clearly defined target customer, use case, value proposition, supported features, and excluded features.
G2	Security and privacy	Threat model, access control, secret management, security tests, privacy controls, and remediation evidence.
G3	Reliability and scale	Measured performance, availability objectives, backups, disaster recovery, observability, and failure testing.
G4	AI safety and quality	Grounded outputs, evaluation metrics, human approval for high-impact actions, model monitoring, and fallback behavior.
G5	Legal and compliance	Licenses, terms, privacy notices, data-processing terms, records, and market-specific compliance review.
G6	Operations and support	Release automation, runbooks, incident response, support ownership, customer onboarding, and rollback readiness.
G7	Commercial readiness	Packaging, pricing, contracts, billing, sales material, implementation plan, and a repeatable pilot offer.
G8	Launch approval	Signed evidence from product, engineering, security, legal, operations, and business owners.

Recommended launch path: Internal alpha -> controlled design-partner pilot -> limited production -> general availability. Advancement should depend on evidence, not only a calendar date.

01**Product and customer readiness**

Prove that the platform solves a specific, valuable problem for a known buyer.

Target market and buyer	Named customer segment, buyer, users, economic problem, and purchase trigger.	TBD
Primary use cases	Three to five production use cases with inputs, outputs, boundaries, and measurable outcomes.	TBD
Release scope	Feature inventory tagged supported, preview, internal-only, deprecated, or excluded.	TBD
User journeys	Validated onboarding, daily workflow, approvals, reporting, administration, and offboarding.	TBD
Value measurement	Baseline and target metrics for time saved, decision quality, risk reduction, revenue, cost, or resilience.	TBD
Accessibility and usability	Keyboard, contrast, readable states, responsive layouts, error messages, and usability test findings.	TBD
Documentation	Product overview, setup guide, admin guide, API guide, user guide, limitations, and release notes.	TBD

Architecture baseline

- Freeze the v1.0 logical architecture, deployment topology, data flows, trust boundaries, integrations, and dependency inventory.
- Document tenant isolation, regional deployment choices, data residency assumptions, supported browsers, supported databases, and supported runtime versions.
- Remove demo-only shortcuts, hard-coded sample values, test credentials, and static intelligence from production paths.
- Define backward-compatibility rules for APIs, database migrations, configuration, and customer integrations.

02

Security, identity, and privacy

Protect enterprise data and establish defensible access and audit controls.

Identity and SSO	Production identity provider integration; MFA/SSO policy; secure session handling.	TBD
Authorization	Role and permission matrix; least privilege; tenant and record-level access tests.	TBD
API security	Authentication, authorization, input validation, rate limits, safe errors, and abuse protection.	TBD
Secrets and keys	No secrets in source control; managed storage; rotation and revocation procedure.	TBD
Encryption	Encryption in transit and at rest; certificate and key ownership documented.	TBD
Auditability	Immutable or protected logs for sign-in, data access, configuration, approvals, AI actions, and exports.	TBD
Secure SDLC	Dependency scanning, static analysis, dynamic testing, container scanning, and remediation workflow.	TBD
Threat model	Threats, abuse cases, trust boundaries, mitigations, residual risks, and owner sign-off.	TBD
Privacy controls	Data inventory, purpose, retention, deletion, export, consent/notice, and processor mapping.	TBD
Penetration test	Independent or qualified test completed; critical/high findings resolved or formally accepted.	TBD

Do not launch if

Cross-tenant access is possible; privileged actions are unaudited; secrets remain in code; critical vulnerabilities are open; or customer data can be used by AI components without an approved data-handling design.

03

AI quality, safety, and governance

Move from generated text to measurable, controlled enterprise intelligence.

Model and provider strategy	Approved models/providers, versions, regions, cost limits, fallback path, and change-control policy.	TBD
Grounding and provenance	Authoritative data sources, citations/provenance, freshness rules, and missing-data behavior.	TBD
Evaluation suite	Representative test set and defined quality, factuality, relevance, safety, and latency measures.	TBD
Human oversight	Approval gates for high-impact recommendations and actions; clear responsibility for final decisions.	TBD
Prompt and output controls	Prompt versioning, injection defenses, output validation, restricted actions, and safe rendering.	TBD
Monitoring	Quality drift, model changes, harmful failures, cost, latency, token usage, and customer feedback.	TBD
Explainability	Users can see the basis, assumptions, confidence/limitations, and source evidence for recommendations.	TBD
Feedback and correction	Users can flag, correct, reject, and trace problematic outputs; fixes enter the evaluation suite.	TBD
Data use boundaries	Customer content, telemetry, retention, model training use, and subprocessors are explicitly controlled.	TBD

Release principle: NexusOne AI should recommend, explain, and request approval before executing consequential enterprise actions. Autonomous execution should be introduced only for narrowly bounded, reversible, monitored actions with tested rollback.

04**Reliability, performance, and recovery**

Prove that the platform remains usable under expected load and recoverable after failure.

Service objectives	Defined availability, latency, error-rate, recovery, and data-loss objectives by critical workflow.	TBD
Load and stress tests	Results for expected and peak concurrency, dataset sizes, API volume, report generation, and AI calls.	TBD
Resilience tests	Timeout, retry, circuit-breaker, queue, dependency outage, partial failure, and degraded-mode evidence.	TBD
Database readiness	Indexes, migrations, transactions, connection pools, query monitoring, archival, and capacity plan.	TBD
Backup and restore	Automated backups plus successful restore test with documented recovery evidence.	TBD
Disaster recovery	Runbook, responsibilities, alternate environment/region decision, and completed recovery exercise.	TBD
Observability	Correlated logs, metrics, traces, dashboards, alerts, health checks, and customer-impact visibility.	TBD
Cost controls	Infrastructure and AI cost budgets, alerts, quotas, rate limits, and per-tenant usage visibility.	TBD

Required proof artifacts

- Performance report with environment, dataset, test scenario, result, bottleneck, and accepted limit.
- Backup/restore and disaster-recovery test record.
- Production dashboard and alert map linked to service objectives.
- Capacity and cost model for pilot, limited production, and scaled operation.

05

Testing, DevSecOps, and operations

Make every release repeatable, observable, reversible, and supportable.

Automated tests	Unit, integration, API contract, UI, security, migration, and regression suites in the pipeline.	TBD
Test data	Synthetic or approved data; masked production-like data; repeatable seed and cleanup process.	TBD
CI/CD	Controlled build, quality gates, artifact signing/provenance, deployment approvals, and environment promotion.	TBD
Configuration	Environment-specific configuration, feature flags, secret references, and validated defaults.	TBD
Release strategy	Blue/green, canary, or controlled rollout; rollback steps and database compatibility tested.	TBD
Incident response	Severity model, on-call contacts, triage, communication, evidence preservation, and post-incident review.	TBD
Runbooks	Top failure modes, dependency outages, data issues, bad AI output, security alert, and rollback.	TBD
Support model	Support channels, ownership, service hours, escalation route, response targets, and known limitations.	TBD
Change management	Release notes, customer notice, maintenance policy, deprecation policy, and emergency-change process.	TBD

Minimum release rule: a production build must be reproducible from source, traceable to a commit, tested automatically, deployed through an approved pipeline, monitored after release, and reversible without improvisation.

06**Legal, compliance, and commercial readiness**

Ensure that NexusOne AI can be bought, deployed, and operated with clear obligations.

Intellectual property	Ownership confirmed for source, generated assets, datasets, models, prompts, and third-party components.	TBD
Open-source licenses	Software bill of materials; license obligations and restricted licenses reviewed.	TBD
Customer agreements	Terms, order form, support terms, warranty/limitation language, and acceptable-use terms reviewed.	TBD
Privacy documents	Privacy notice, data-processing terms, subprocessor list, retention/deletion commitments.	TBD
Compliance scope	Applicable industry and market requirements identified; claims backed by evidence.	TBD
AI transparency	Customer-facing description of AI use, limitations, oversight, data handling, and model/provider changes.	TBD
Packaging and pricing	Edition/plan boundaries, usage meters, limits, implementation fees, and commercial approvals.	TBD
Billing and entitlement	Subscription, licensing, tenant provisioning, renewal, suspension, and offboarding tested.	TBD
Sales readiness	Qualified pitch, demo script, security FAQ, architecture pack, ROI narrative, and pilot proposal.	TBD

Important

Legal and regulatory requirements vary by customer, industry, and country. This guide is a product-planning checklist, not legal advice. Obtain qualified review before making compliance, safety, availability, or performance claims.

07**Pilot, launch, and acceptance plan**

Convert technical readiness into customer evidence through controlled stages.

Internal alpha	Employees and test tenants	Core journeys work; no critical security defects; telemetry and feedback active.
Design-partner pilot	Selected customer and bounded use case	Data agreement signed; success measures defined; human approval enabled; support channel active.
Limited production	Small controlled customer set	Operational objectives met; restore and rollback tested; incident process proven; commercial terms active.
General availability	Approved market segment	All release gates signed; support and scaling capacity confirmed; documentation and contracts published.

Pilot success measures

- Adoption: invited users complete the intended workflow without assisted workarounds.
- Value: the agreed business outcome improves versus the starting baseline.
- Quality: AI and analytics outputs meet the agreed acceptance threshold on representative work.
- Reliability: customer-impacting errors, latency, and downtime remain within pilot objectives.
- Safety: high-impact decisions remain reviewable, attributable, and reversible.
- Economics: infrastructure, model, implementation, and support costs fit the intended commercial model.

Go / No-Go decision

A release is GO only when every mandatory gate has evidence, every blocker has an owner and resolution, residual risks are explicitly accepted by authorized owners, and rollback/support capacity is ready. Otherwise the release remains NO-GO or limited to a controlled pilot.

Master Release Sign-Off

Complete this page immediately before each external production release.

Product	Scope, target customer, supported journeys, limitations, and acceptance criteria approved.			GO / NO-GO
Engineering	Build, architecture, migrations, integrations, performance, rollback, and technical debt reviewed.			GO / NO-GO
Quality	Regression, security, usability, accessibility, recovery, and release-candidate tests passed.			GO / NO-GO
Security	Threat model, vulnerabilities, access controls, audit, secrets, and penetration-test findings accepted.			GO / NO-GO
Privacy / Legal	Data handling, contracts, notices, licenses, subprocessors, and market obligations approved.			GO / NO-GO
AI Governance	Models, data, evaluation, oversight, monitoring, change control, and fallback approved.			GO / NO-GO
Operations	Monitoring, alerts, runbooks, on-call, incident process, backup/restore, and support ready.			GO / NO-GO
Commercial	Pricing, entitlement, billing, sales materials, pilot/implementation plan, and customer terms ready.			GO / NO-GO
Executive sponsor	Residual risks accepted and final launch decision recorded.			GO / NO-GO

Current roadmap position	Completed locally through Phase 9.34 - Investment Strategy. Planned next: Phase 9.35 through Phase 9.44. Feature completion and production readiness should be tracked as separate percentages.
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Suggested repository location: docs/NexusOne_AI_Pre_Market_Release_Readiness_Guide.pdf